

Sealed Experiment Record

Sealed 2026-10-07 · Signed by the K-Veritas server
Session 2ac306cc · Data hash f549...597e

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-07 08:41 and 09:01 UTC. Reported results: linear_eval_test_top1 63.0314, linear_eval_test_top5 96.3971. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	2ac306cc-aeaa-4362-bf2d-7f4ff05f4beb
Initialized	2026-10-07T08:41:16Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	3289d1db6ae574b99b8cd94eb8c1871bb121b518057eed45bb7c70252fb9803c
Disclosure	open
Source code, run 1	477ab23b0c6dd571455d4a35868e9b0a07588629ea719ea47f9feb468a50c687

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	20 minutes 38 secon...	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
contrastive_loss	2.46374	49 (50 readings)	explicit
contrastive_top1	50	49 (50 readings)	explicit
linear_train_top1	66.0738	99 (100 readings)	explicit
linear_test_top1	63.0314	99 (100 readings)	explicit

Inline claim	Value	Phase	Output line
linear_eval_test_top1	63.0314		467
linear_eval_test_top5	96.3971		468

Seed committed: seed:0, output line 1, 2026-10-07T08:41:16Z.

4 Execution evidence

Coherence

Coherent (0.40): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	7200	494.1 s	30.75 GB	64%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 12 tracked files, 4 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	08:41:16	010abfffd9e31a5ac	12
3	run_end	09:01:54	010abfffd9e31a5ac	0

2 snapshots with no file change omitted here; full timeline in Appendix B.

Activity: 12 files read, 3 files written, 1603 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-07T09:01:56.728066+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	f549474966292babd57d12e269d8acc3b6077e6c111df2ade35d5815ede9597e
Code bundle hash	719c811c6120a0fcdda61c9028456c513c84eb9ce99ffc9e87022e095d0d52c1

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
contrastive_loss	5.38849	0	14	explicit
contrastive_top1	1.5625	0	15	explicit
contrastive_loss	5.10693	1	17	explicit
contrastive_top1	9.17969	1	18	explicit
contrastive_loss	4.78898	2	20	explicit
contrastive_top1	13.4766	2	21	explicit
contrastive_loss	4.64238	3	23	explicit
contrastive_top1	16.0156	3	24	explicit
contrastive_loss	4.14925	4	26	explicit
contrastive_top1	20.5078	4	27	explicit
contrastive_loss	4.25722	5	29	explicit
contrastive_top1	19.5312	5	30	explicit
contrastive_loss	4.38657	6	32	explicit
contrastive_top1	27.7344	6	33	explicit
contrastive_loss	3.96652	7	35	explicit
contrastive_top1	26.3672	7	36	explicit
contrastive_loss	3.84898	8	38	explicit
contrastive_top1	28.5156	8	39	explicit
contrastive_loss	3.68056	9	41	explicit
contrastive_top1	28.3203	9	42	explicit
contrastive_loss	3.76875	10	44	explicit
contrastive_top1	34.375	10	45	explicit
contrastive_loss	3.8403	11	47	explicit
contrastive_top1	22.4609	11	48	explicit
contrastive_loss	3.64376	12	50	explicit
contrastive_top1	32.4219	12	51	explicit
contrastive_loss	3.43282	13	53	explicit
contrastive_top1	33.9844	13	54	explicit
contrastive_loss	3.65985	14	56	explicit
contrastive_top1	34.375	14	57	explicit
contrastive_loss	3.47282	15	59	explicit
contrastive_top1	37.3047	15	60	explicit
contrastive_loss	3.48642	16	62	explicit
contrastive_top1	31.6406	16	63	explicit
contrastive_loss	3.19181	17	65	explicit
contrastive_top1	40.625	17	66	explicit
contrastive_loss	3.17813	18	68	explicit
contrastive_top1	38.6719	18	69	explicit
contrastive_loss	3.2666	19	71	explicit
contrastive_top1	38.8672	19	72	explicit
contrastive_loss	3.06335	20	74	explicit
contrastive_top1	36.1328	20	75	explicit
contrastive_loss	2.99898	21	77	explicit
contrastive_top1	34.9609	21	78	explicit

Metric	Value	Step	Line	Source
contrastive_loss	2.80227	22	80	explicit
contrastive_top1	43.75	22	81	explicit
contrastive_loss	2.92449	23	83	explicit
contrastive_top1	38.8672	23	84	explicit
contrastive_loss	2.71601	24	86	explicit
contrastive_top1	44.5312	24	87	explicit
contrastive_loss	2.89768	25	89	explicit
contrastive_top1	45.7031	25	90	explicit
contrastive_loss	3.07329	26	92	explicit
contrastive_top1	39.6484	26	93	explicit
contrastive_loss	3.31496	27	95	explicit
contrastive_top1	44.9219	27	96	explicit
contrastive_loss	3.33687	28	98	explicit
contrastive_top1	43.3594	28	99	explicit
contrastive_loss	3.08613	29	101	explicit
contrastive_top1	42.5781	29	102	explicit
contrastive_loss	2.79018	30	104	explicit
contrastive_top1	44.9219	30	105	explicit
contrastive_loss	2.47984	31	107	explicit
contrastive_top1	40.0391	31	108	explicit
contrastive_loss	2.92498	32	110	explicit
contrastive_top1	45.7031	32	111	explicit
contrastive_loss	2.78659	33	113	explicit
contrastive_top1	45.5078	33	114	explicit
contrastive_loss	2.91588	34	116	explicit
contrastive_top1	43.1641	34	117	explicit
contrastive_loss	3.07253	35	119	explicit
contrastive_top1	49.6094	35	120	explicit
contrastive_loss	2.67224	36	122	explicit
contrastive_top1	42.3828	36	123	explicit
contrastive_loss	2.71081	37	125	explicit
contrastive_top1	45.5078	37	126	explicit
contrastive_loss	2.49839	38	128	explicit
contrastive_top1	49.0234	38	129	explicit
contrastive_loss	3.02672	39	131	explicit
contrastive_top1	50.5859	39	132	explicit
contrastive_loss	2.38074	40	134	explicit
contrastive_top1	50.9766	40	135	explicit
contrastive_loss	2.32672	41	137	explicit
contrastive_top1	48.8281	41	138	explicit
contrastive_loss	2.75248	42	140	explicit
contrastive_top1	52.1484	42	141	explicit
contrastive_loss	2.78043	43	143	explicit
contrastive_top1	46.875	43	144	explicit
contrastive_loss	2.76853	44	146	explicit
contrastive_top1	51.7578	44	147	explicit
contrastive_loss	2.78436	45	149	explicit
contrastive_top1	54.6875	45	150	explicit
contrastive_loss	2.47804	46	152	explicit
contrastive_top1	57.4219	46	153	explicit

Metric	Value	Step	Line	Source
contrastive_loss	2.44272	47	155	explicit
contrastive_top1	49.6094	47	156	explicit
contrastive_loss	2.26262	48	158	explicit
contrastive_top1	57.0312	48	159	explicit
contrastive_loss	2.46374	49	161	explicit
contrastive_top1	50	49	162	explicit
linear_train_top1	48.1413	0	168	explicit
linear_test_top1	56.1811	0	169	explicit
linear_train_top1	58.2514	1	171	explicit
linear_test_top1	58.2014	1	172	explicit
linear_train_top1	59.7202	2	174	explicit
linear_test_top1	59.325	2	175	explicit
linear_train_top1	60.6409	3	177	explicit
linear_test_top1	59.8231	3	178	explicit
linear_train_top1	61.2907	4	180	explicit
linear_test_top1	60.3372	4	181	explicit
linear_train_top1	61.6996	5	183	explicit
linear_test_top1	60.6376	5	184	explicit
linear_train_top1	62.0942	6	186	explicit
linear_test_top1	60.9013	6	187	explicit
linear_train_top1	62.4099	7	189	explicit
linear_test_top1	61.1868	7	190	explicit
linear_train_top1	62.675	8	192	explicit
linear_test_top1	61.3528	8	193	explicit
linear_train_top1	62.9337	9	195	explicit
linear_test_top1	61.5079	9	196	explicit
linear_train_top1	63.0931	10	198	explicit
linear_test_top1	61.6751	10	199	explicit
linear_train_top1	63.2932	11	201	explicit
linear_test_top1	61.8325	11	202	explicit
linear_train_top1	63.5009	12	204	explicit
linear_test_top1	62.0657	12	205	explicit
linear_train_top1	63.6507	13	207	explicit
linear_test_top1	62.222	13	208	explicit
linear_train_top1	63.7584	14	210	explicit
linear_test_top1	62.3001	14	211	explicit
linear_train_top1	63.8879	15	213	explicit
linear_test_top1	62.2524	15	214	explicit
linear_train_top1	63.9756	16	216	explicit
linear_test_top1	62.2622	16	217	explicit
linear_train_top1	64.0812	17	219	explicit
linear_test_top1	62.2915	17	220	explicit
linear_train_top1	64.1765	18	222	explicit
linear_test_top1	62.3696	18	223	explicit
linear_train_top1	64.2363	19	225	explicit
linear_test_top1	62.4575	19	226	explicit
linear_train_top1	64.2941	20	228	explicit
linear_test_top1	62.5161	20	229	explicit
linear_train_top1	64.4077	21	231	explicit
linear_test_top1	62.5356	21	232	explicit

Metric	Value	Step	Line	Source
linear_train_top1	64.4814	22	234	explicit
linear_test_top1	62.477	22	235	explicit
linear_train_top1	64.5297	23	237	explicit
linear_test_top1	62.5345	23	238	explicit
linear_train_top1	64.5894	24	240	explicit
linear_test_top1	62.5931	24	241	explicit
linear_train_top1	64.6751	25	243	explicit
linear_test_top1	62.5758	25	244	explicit
linear_train_top1	64.7409	26	246	explicit
linear_test_top1	62.5747	26	247	explicit
linear_train_top1	64.7628	27	249	explicit
linear_test_top1	62.5454	27	250	explicit
linear_train_top1	64.8007	28	252	explicit
linear_test_top1	62.5747	28	253	explicit
linear_train_top1	64.8629	29	255	explicit
linear_test_top1	62.6626	29	256	explicit
linear_train_top1	64.9286	30	258	explicit
linear_test_top1	62.7212	30	259	explicit
linear_train_top1	64.9685	31	261	explicit
linear_test_top1	62.7407	31	262	explicit
linear_train_top1	65.0323	32	264	explicit
linear_test_top1	62.7798	32	265	explicit
linear_train_top1	65.0721	33	267	explicit
linear_test_top1	62.8286	33	268	explicit
linear_train_top1	65.104	34	270	explicit
linear_test_top1	62.8177	34	271	explicit
linear_train_top1	65.1519	35	273	explicit
linear_test_top1	62.8665	35	274	explicit
linear_train_top1	65.1961	36	276	explicit
linear_test_top1	62.847	36	277	explicit
linear_train_top1	65.22	37	279	explicit
linear_test_top1	62.8556	37	280	explicit
linear_train_top1	65.2679	38	282	explicit
linear_test_top1	62.8556	38	283	explicit
linear_train_top1	65.2838	39	285	explicit
linear_test_top1	62.8263	39	286	explicit
linear_train_top1	65.3117	40	288	explicit
linear_test_top1	62.7872	40	289	explicit
linear_train_top1	65.3217	41	291	explicit
linear_test_top1	62.7786	41	292	explicit
linear_train_top1	65.3396	42	294	explicit
linear_test_top1	62.8567	42	295	explicit
linear_train_top1	65.3834	43	297	explicit
linear_test_top1	62.8763	43	298	explicit
linear_train_top1	65.3834	44	300	explicit
linear_test_top1	62.8958	44	301	explicit
linear_train_top1	65.3954	45	303	explicit
linear_test_top1	62.8958	45	304	explicit
linear_train_top1	65.4153	46	306	explicit
linear_test_top1	62.9349	46	307	explicit

Metric	Value	Step	Line	Source
linear_train_top1	65.4452	47	309	explicit
linear_test_top1	62.9642	47	310	explicit
linear_train_top1	65.4691	48	312	explicit
linear_test_top1	63.0032	48	313	explicit
linear_train_top1	65.513	49	315	explicit
linear_test_top1	62.9837	49	316	explicit
linear_train_top1	65.5349	50	318	explicit
linear_test_top1	62.9446	50	319	explicit
linear_train_top1	65.5628	51	321	explicit
linear_test_top1	62.9544	51	322	explicit
linear_train_top1	65.5871	52	324	explicit
linear_test_top1	62.9642	52	325	explicit
linear_train_top1	65.5951	53	327	explicit
linear_test_top1	62.9349	53	328	explicit
linear_train_top1	65.617	54	330	explicit
linear_test_top1	62.9153	54	331	explicit
linear_train_top1	65.6469	55	333	explicit
linear_test_top1	62.8958	55	334	explicit
linear_train_top1	65.6633	56	336	explicit
linear_test_top1	62.9446	56	337	explicit
linear_train_top1	65.6952	57	339	explicit
linear_test_top1	62.9739	57	340	explicit
linear_train_top1	65.7171	58	342	explicit
linear_test_top1	62.9739	58	343	explicit
linear_train_top1	65.745	59	345	explicit
linear_test_top1	62.9837	59	346	explicit
linear_train_top1	65.7549	60	348	explicit
linear_test_top1	62.9837	60	349	explicit
linear_train_top1	65.7828	61	351	explicit
linear_test_top1	62.9739	61	352	explicit
linear_train_top1	65.7769	62	354	explicit
linear_test_top1	62.9739	62	355	explicit
linear_train_top1	65.7709	63	357	explicit
linear_test_top1	62.9642	63	358	explicit
linear_train_top1	65.7809	64	360	explicit
linear_test_top1	62.9544	64	361	explicit
linear_train_top1	65.7928	65	363	explicit
linear_test_top1	62.9837	65	364	explicit
linear_train_top1	65.8008	66	366	explicit
linear_test_top1	62.9923	66	367	explicit
linear_train_top1	65.8028	67	369	explicit
linear_test_top1	63.0021	67	370	explicit
linear_train_top1	65.8147	68	372	explicit
linear_test_top1	62.9825	68	373	explicit
linear_train_top1	65.8287	69	375	explicit
linear_test_top1	62.9912	69	376	explicit
linear_train_top1	65.8546	70	378	explicit
linear_test_top1	62.9716	70	379	explicit
linear_train_top1	65.8546	71	381	explicit
linear_test_top1	62.9521	71	382	explicit

Metric	Value	Step	Line	Source
linear_train_top1	65.8626	72	384	explicit
linear_test_top1	62.9619	72	385	explicit
linear_train_top1	65.8546	73	387	explicit
linear_test_top1	62.9716	73	388	explicit
linear_train_top1	65.8646	74	390	explicit
linear_test_top1	62.9912	74	391	explicit
linear_train_top1	65.8845	75	393	explicit
linear_test_top1	63.0009	75	394	explicit
linear_train_top1	65.8984	76	396	explicit
linear_test_top1	63.0009	76	397	explicit
linear_train_top1	65.9104	77	399	explicit
linear_test_top1	63.0107	77	400	explicit
linear_train_top1	65.9283	78	402	explicit
linear_test_top1	63.0205	78	403	explicit
linear_train_top1	65.9483	79	405	explicit
linear_test_top1	63.0595	79	406	explicit
linear_train_top1	65.9503	80	408	explicit
linear_test_top1	63.0302	80	409	explicit
linear_train_top1	65.9582	81	411	explicit
linear_test_top1	63.0205	81	412	explicit
linear_train_top1	65.9662	82	414	explicit
linear_test_top1	63.0302	82	415	explicit
linear_train_top1	65.9662	83	417	explicit
linear_test_top1	63.0693	83	418	explicit
linear_train_top1	65.9503	84	420	explicit
linear_test_top1	63.0693	84	421	explicit
linear_train_top1	65.9443	85	423	explicit
linear_test_top1	63.0693	85	424	explicit
linear_train_top1	65.9602	86	426	explicit
linear_test_top1	63.0498	86	427	explicit
linear_train_top1	65.9642	87	429	explicit
linear_test_top1	63.0595	87	430	explicit
linear_train_top1	65.9782	88	432	explicit
linear_test_top1	63.0888	88	433	explicit
linear_train_top1	65.9881	89	435	explicit
linear_test_top1	63.079	89	436	explicit
linear_train_top1	66.0001	90	438	explicit
linear_test_top1	63.0693	90	439	explicit
linear_train_top1	66.014	91	441	explicit
linear_test_top1	63.04	91	442	explicit
linear_train_top1	66.014	92	444	explicit
linear_test_top1	63.0009	92	445	explicit
linear_train_top1	66.026	93	447	explicit
linear_test_top1	63.0302	93	448	explicit
linear_train_top1	66.0379	94	450	explicit
linear_test_top1	63.0107	94	451	explicit
linear_train_top1	66.0539	95	453	explicit
linear_test_top1	63.04	95	454	explicit
linear_train_top1	66.0619	96	456	explicit
linear_test_top1	63.04	96	457	explicit

[illegible]

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[illegible]

[illegible]

[illegible]

[illegible]

D Raw seal fields and hashes

Seal

```
data_hash          f549474966292babd57d12e269d8acc3b6077e6c111df2ade35d5815ede9597e
signed_message_hash a11770b37e724eb73d6a61add2f898c226b279ffceebd8cc8062854435b9d078
nonce             5ea7c379b3698c7658729c015f1688e9
signed_at         2026-10-07T09:01:56.728066+00:00
sealed_at         2026-10-07T09:01:56.786884397Z
checkout_bundle   719c811c6120a0fcdda61c9028456c513c84eb9ce99ffc9e87022e095d0d52c1
```

Signature (base64)

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LRGtKIqBGh0+Abb3RP05RREA55iMXGtLXC07FpxVf7Skqx3cIv0w3eZRj9dZleSDktY=
```

Public key

```
-----BEGIN PUBLIC KEY-----
MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEA1QL3YyMgeShx8IhTC7jb
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7qU95EFEEisX/GfoS9ToxtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KICeAdbiHN
av0JjdTJ//CM0WZSIbdzE6qvEbNfcqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdHQ+D
AhfTRuJNgWEkjwD5lb0Vhi4qlkweSOEMTKT3ye9IL5xEaJn4bku4eKqes7VSZt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPgPZ242NH8JgE6CNF7ToL9syans67u7UMgIcNgeNHnCy9bLgPU5annzP9gN2l
6HEMLdkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7LE3mbQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCKDxoehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PWjNcNku5cdW3gHLcCAwEAAQ==
-----END PUBLIC KEY-----
```

Run 1 outputs

```
stdout 72c137af974770d3c7490076423a79433430ef8c850939c82bd944ecba66b383
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics cfc71ed83096a4af84f4f0563c0b93e98dadb23402d1f4e34398cb611e9af183
digest c4ad8b5b54b42bff876bda4d5c266ba7814591b253c585f751287f8facf606c9
kv_run.sh (unchanged)
pre 2ab560ccb47fe19fd373dca3df3b48baf1e41f4daabe74a323228765b8f287a2
post 2ab560ccb47fe19fd373dca3df3b48baf1e41f4daabe74a323228765b8f287a2
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}
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